

Acute Leukemia Classification Using Gene Expression

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Abstract

Accurate differentiation between acute myeloid leukemia (AML) and acute lymphoblastic leukemia (ALL) is clinically important because the two diseases arise from different blood-cell lineages and require different treatment strategies. This study evaluates whether gene-expression profiles can distinguish AML from ALL in the benchmark Golub et al. microarray dataset, which contains 7,129 probes measured in 72 leukemia samples. Using the original split of 38 development samples and 34 independent test samples, three supervised learning models were compared: a decision tree, logistic regression, and a linear support vector machine (SVM).

On the independent test set, the decision tree and SVM each correctly classified 31 of 34 patients (91.18% accuracy), while logistic regression correctly classified 28 of 34 patients (82.35%). Although the decision tree and SVM achieved the same overall accuracy, they produced different error patterns: the decision tree achieved the highest AML recall, whereas the SVM produced no false-positive AML predictions. Logistic regression was more conservative and missed more AML cases. Feature-ranking analyses from logistic regression and the SVM showed strong agreement, with 16 of their top 20 probes overlapping. Probe X70297 (annotated as *CHRNA7*) and probe X95735 (*ZYX*/zyxin) emerged as especially influential features.

Overall, the results show that gene-expression data contain a strong signal for separating AML from ALL, while also highlighting the importance of class-specific error patterns, small-sample limitations, and the need for independent validation before clinical interpretation.

Keywords: acute myeloid leukemia; acute lymphoblastic leukemia; gene expression; microarray; decision tree; logistic regression; support vector machine; biomarker discovery

1 Introduction

Leukemia is a group of cancers that begin in blood-forming tissues, most often the bone marrow. The disease disrupts normal blood-cell development and can reduce the production of functional red blood cells, platelets, and white blood cells. Clinicians classify leukemia along two main dimensions: whether the disease progresses rapidly (acute) or more slowly (chronic), and whether it arises from lymphoid or myeloid cell lineages [1]. This project focuses on the two major acute categories, acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML).

Distinguishing AML from ALL matters because the two diseases differ biologically and are treated differently. In clinical practice, diagnosis does not rely on a single blood-cell count or a machine-learning prediction. Instead, physicians integrate a complete blood count, examination of blood and bone-marrow cells, immunophenotyping, cytogenetic testing, and molecular analysis [2, 3]. Gene-expression profiling can add another layer of information by measuring the activity of thousands of genes simultaneously. The central question of this project is whether these high-dimensional molecular patterns contain enough structure to classify new leukemia samples accurately.

The project addresses four research questions:

1. How accurately do a decision tree, logistic regression, and a linear SVM distinguish AML from ALL on an independent test set?
2. Do the models make the same types of errors, or do they show different trade-offs between sensitivity and precision?
3. Which gene-expression probes contribute most strongly to the linear classification boundaries?
4. Are the leading probes consistent across models and with findings reported in previous analyses of the Golub dataset?

The goal is not merely to report three accuracy values. A strong classification analysis must

explain the dataset, show how the models were evaluated, interpret the confusion matrices, distinguish exploratory findings from validated findings, and acknowledge the limitations created by a dataset with thousands of predictors but only dozens of patients.

2 Background and Literature Review

2.1 Leukemia categories and diagnosis

The four commonly discussed leukemia categories are summarized in Table 1. The acute forms involve immature precursor cells and generally require prompt evaluation and treatment. The chronic forms usually progress more slowly, although their clinical courses vary considerably. This biological framework is important because the machine-learning task in this study is not “leukemia versus no leukemia”; it is the more specific problem of separating two acute leukemias that arise from different developmental lineages.

Table 1: High-level comparison of the four major leukemia categories.

Type	Course	Primary lineage	Typical pattern	Diagnostic context
ALL	Acute	Lymphoid	Common in children but also occurs in adults; characterized by lymphoblast proliferation	Blood and bone-marrow evaluation, flow cytometry, and genetic testing help establish lineage and subtype
AML	Acute	Myeloid	More common in adults, especially older adults, but may occur at any age; characterized by myeloid blasts	Morphology, immunophenotyping, cytogenetics, and molecular findings guide diagnosis and risk classification
CLL	Chronic	Lymphoid	Usually affects adults and may progress slowly over years	Often identified through persistent lymphocytosis and immunophenotyping
CML	Chronic	Myeloid	Usually affects adults; may be discovered before severe symptoms develop	Strongly associated with the BCR::ABL1 fusion and evaluated using molecular and cytogenetic tests

A complete blood count can reveal abnormal cell numbers, but an abnormal white-blood-cell count alone cannot determine leukemia subtype. Bone-marrow examination provides direct information about blast cells, while flow cytometry identifies lineage-specific surface markers. Cytogenetic and molecular tests provide additional information for diagnosis, prognosis, and treatment selection [2, 3]. Consequently, this study evaluates whether expression-based models can recognize molecular class patterns; it does not propose replacing standard clinical testing.

2.2 The Golub gene-expression study

Golub et al. introduced one of the earliest and most influential demonstrations that cancer classes could be predicted from genome-wide expression patterns [4]. Their work used DNA microarrays to measure thousands of gene-expression values in acute leukemia samples. The study showed that an automatically derived molecular predictor could distinguish AML from ALL in previously unseen cases, providing an early example of data-driven precision oncology.

The version of the dataset used in this project contains 72 samples and 7,129 probe measurements. Importantly, the class counts are 47 ALL and 25 AML, not the reverse. The 38-sample development set contains 27 ALL and 11 AML samples, while the independent 34-sample test set contains 20 ALL and 14 AML samples. This preserved split makes the test set more informative than a training accuracy because the test patients were not used to fit the model parameters.

The Golub dataset became a benchmark for statistical learning in genomics. Dudoit, Fridlyand, and Speed compared multiple discrimination methods, including classification trees, on leukemia and other tumor-expression datasets [5]. Furey et al. demonstrated the usefulness of SVMs for high-dimensional microarray classification [7]. Li and Yang showed that a small number of genes could carry strong discriminative information in the Golub data and reported zyxin (probe X95735) as an especially strong single-gene predictor [8]. Later work continued to identify zyxin and other compact gene sets in leukemia-classification analyses [9]. These studies support the value of comparing models while also warning that apparent performance can be overly optimistic if feature selection and evaluation are not separated correctly.

2.3 Why compare three models?

The three selected models represent different approaches to classification.

Decision tree. A decision tree divides the feature space through a sequence of rules. At each node,

the algorithm chooses a feature and cutoff that improve class separation. Trees are easy to interpret because a prediction can be traced through explicit decisions. They also capture nonlinear interactions without requiring standardized predictors. Their main weakness is instability: with a small dataset, a slight change in samples can produce a different tree, and a deep tree can overfit.

Logistic regression. Logistic regression models the probability of AML through a weighted linear combination of gene-expression values. For a patient with feature vector $x = (x_1, \dots, x_p)$,

$$P(Y = 1 | x) = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)]}, \quad (1)$$

where $Y = 1$ represents AML and $Y = 0$ represents ALL. The sign and magnitude of β_j describe how a standardized feature contributes to the fitted classification boundary. Because this dataset has many more predictors than training samples, regularization is essential to control coefficient size and reduce overfitting.

Support vector machine. A linear SVM searches for a separating hyperplane

$$w^\top x + b = 0 \quad (2)$$

that creates a large margin between classes while allowing some classification errors. Samples are assigned according to the sign of $w^\top x + b$. SVMs are well suited to high-dimensional settings because the decision boundary depends primarily on the samples closest to the margin, called support vectors [6]. However, SVM scores are not automatically calibrated probabilities, and the feature weights should be interpreted cautiously when predictors are strongly correlated.

Table 2: Reasons for including each model in the comparison.

Model	Main advantage for this project	Primary risk
Decision tree	Produces explicit threshold rules and may capture nonlinear structure	High variance and possible overfitting in a small sample
Logistic regression	Provides probabilistic interpretation and signed coefficients	Requires regularization when $p \gg n$; correlated genes can destabilize coefficients
Linear SVM	Often performs well in high-dimensional, low-sample-size classification	Margin scores are not probabilities; weights are not causal effects

3 Data and Methods

3.1 Dataset structure and outcome definition

Each row in the analytic dataset represents one patient sample, and each gene-expression column represents one microarray probe. The outcome was encoded as AML = 1 and ALL = 0. Table 3 shows the original development and test cohorts.

Table 3: Class distribution in the Golub dataset used for this project.

Cohort	ALL	AML	Total	Purpose
Development set	27	11	38	Model fitting and exploratory gene analysis
Independent test set	20	14	34	Final comparison of predictive performance
Total	47	25	72	Complete benchmark dataset

The central statistical challenge is the extreme ratio of predictors to observations: $p = 7,129$ expression probes but only $n = 38$ development samples. This $p \gg n$ setting allows flexible models to

fit noise unless complexity is controlled and evaluation is performed on data that were not used during model development.

3.2 Preprocessing and model fitting

The analysis followed a common supervised-learning workflow. Sample identifiers and class labels were separated from the gene-expression matrix. The original development/test split was retained rather than randomly reshuffling all 72 patients. For logistic regression and the linear SVM, expression features were standardized. In a leakage-free implementation, the means and standard deviations must be estimated from the development set only and then applied unchanged to the test set. Standardization places probes on comparable scales and prevents a feature with a large numeric range from dominating solely because of its units. A decision tree does not require feature scaling because its rules depend on ordered cutoffs rather than distances.

The three models were fitted on the 38 development samples and evaluated on the 34 independent test samples. The classification threshold for logistic regression was 0.5: probabilities above 0.5 were labeled AML, while lower probabilities were labeled ALL. The SVM assigned labels according to the sign of the decision function. The decision tree followed its learned sequence of feature thresholds.

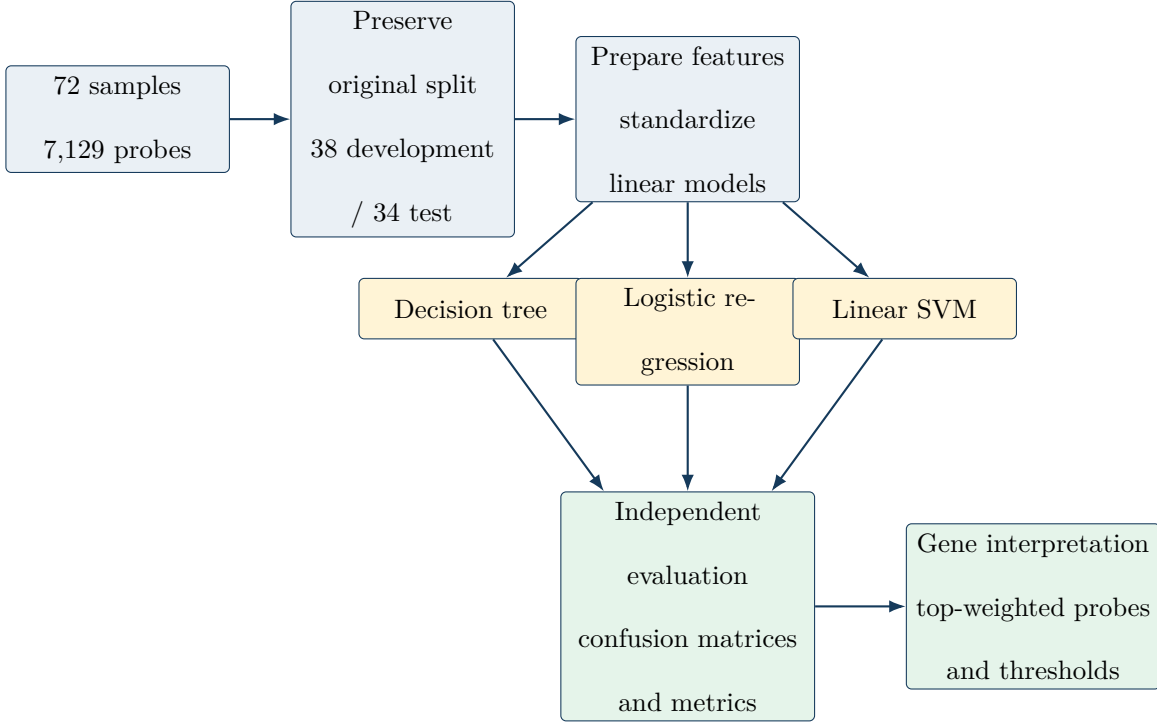


Figure 1: Analysis workflow used to compare models and interpret influential probes.

3.3 Evaluation metrics

Because a model can achieve a reasonable accuracy while repeatedly missing one class, performance was evaluated using the full confusion matrix. With AML treated as the positive class,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (5)$$

$$F_1 = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}}. \quad (6)$$

AML recall is especially important in this project because it measures the proportion of true AML patients recognized by the classifier. Precision answers a different question: among samples predicted as AML, how many truly are AML? Balanced accuracy, the average of the two class-specific recalls, was

also calculated to reduce the influence of unequal class sizes. Approximate 95% Wilson confidence intervals were added for overall accuracy to emphasize the uncertainty created by a test set of only 34 patients.

3.4 Feature ranking and exploratory thresholds

For logistic regression, probes were ranked by the absolute magnitude of their fitted coefficients. For the linear SVM, probes were ranked by the absolute magnitude of their weights. Absolute values were used because both large positive and large negative weights can strongly influence a classification boundary. Agreement between the two top-20 lists was examined as a simple measure of cross-model consistency.

The development set was also used for an exploratory single-gene threshold analysis of *zyxin* (X95735) and *CHRNA7* (X70297). Candidate cutoffs were compared by the number of development samples correctly separated. These thresholds are descriptive, not independently validated: choosing the best cutoff and measuring its accuracy on the same 38 samples produces an optimistically biased estimate. A future study should select the threshold in one cohort and test it unchanged in another.

4 Results

Three models were trained on the same 38 development samples and tested on the same 34 independent samples. Overall performance ranged from 82.35% to 91.18%, but the confusion matrices reveal meaningful differences in how the models handled AML and ALL.

4.1 Overall model performance

Table 4: Overall performance on the 34-sample independent test set.

Model	Correct	Accuracy	95% CI	Macro F_1	Balanced accuracy
Decision tree	31/34	91.18%	77.04–96.95%	0.91	91.43%
Logistic regression	28/34	82.35%	66.49–91.65%	0.80	78.57%
Linear SVM	31/34	91.18%	77.04–96.95%	0.91	89.29%

The decision tree and SVM tied in overall accuracy, but the decision tree had the highest balanced accuracy because it recovered a larger fraction of AML cases. The confidence intervals are wide, demonstrating that 91.18% should not be interpreted as a precise estimate of performance in a broader patient population.

Table 5: Class-specific results and error patterns.

Model	AML precision	AML recall	AML F_1	ALL recall	Observed error pattern
Decision tree	0.87	0.93	0.90	0.90	Missed 1 of 14 AML cases and labeled 2 of 20 ALL cases as AML
Logistic regression	1.00	0.57	0.73	1.00	Made no false-positive AML calls, but missed 6 of 14 AML cases
Linear SVM	1.00	0.79	0.88	1.00	Made no false-positive AML calls and missed 3 of 14 AML cases

4.2 Confusion-matrix interpretation

Decision Tree			Logistic Regression			Linear SVM			
Actual	Predicted		Actual	Predicted		Actual	Predicted		
	ALL	AML		ALL	AML		ALL	AML	
	ALL	18		2	20		0	20	0
	AML	1		13	6		8	3	11

Figure 2: Confusion matrices on the independent test set. Rows are actual classes and columns are predicted classes.

The decision tree correctly identified 13 of 14 AML patients, giving an AML false-negative rate of only 7.14%. Its trade-off was two false-positive AML predictions. Logistic regression had perfect AML precision because all eight predicted AML cases were truly AML, but its recall was only 57.14%; six AML patients were assigned to ALL. The SVM provided a better balance than logistic regression: it retained perfect AML precision while recovering 11 of 14 AML cases.

These results demonstrate why the phrase “100% accuracy for ALL” can be misleading. Logistic regression and the SVM both had ALL recall of 1.00, meaning that every actual ALL sample was recognized. However, logistic regression predicted ALL for six patients who actually had AML. A complete interpretation therefore requires both directions of error.

4.3 Consistency of influential probes

The top-20 logistic-regression and SVM rankings showed substantial agreement. Sixteen probes appeared in both lists, corresponding to an overlap of 80% relative to either 20-probe list and a Jaccard similarity of $16/24 = 0.67$. This consistency suggests that the two linear models captured a shared expression signal rather than producing completely unrelated rankings. It does not, however, prove that every high-weight probe is a causal driver; correlated genes can receive interchangeable weights.

X70297 (*CHRNA7*) ranked first in both models. X95735 (*ZYX*, zyxin) also appeared in both lists. Other consensus probes included M23197 (*CD33*) and M19507 (*MPO*), genes associated with myeloid differentiation and therefore biologically compatible with an AML versus ALL classification task [15, 16]. The appearance of lineage-related probes strengthens the interpretation that the models learned genuine biological structure, although independent replication is still necessary.

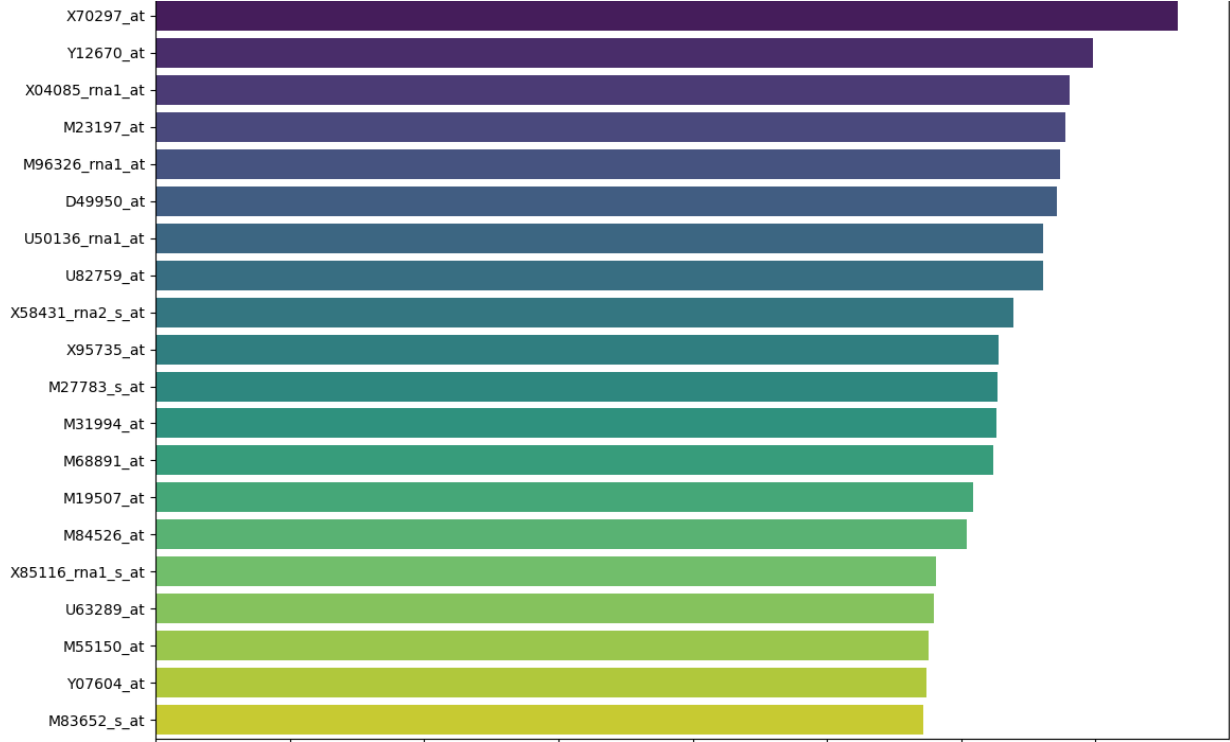


Figure 3: Top 20 probes ranked by absolute logistic-regression coefficient. Larger bars indicate a stronger contribution to the fitted linear decision boundary; direction is not shown because absolute values are plotted.



Figure 4: Top 20 probes ranked by linear-SVM weight. Positive and negative weights can both be influential, so interpretation should consider both magnitude and direction.

4.4 Exploratory analysis of zyxin and *CHRNA7*

Zyxin is a focal-adhesion and cytoskeletal protein involved in mechanosensitive signaling and actin organization [12, 13]. Its repeated appearance in studies of the Golub dataset is notable: Li and Yang reported zyxin as the best single-gene model in their analysis, and Tang et al. also identified X95735 among the most informative leukemia probes [8, 9]. In this project’s 38-sample development set, mean zyxin expression was approximately 3,024 for AML and 350 for ALL. The observed ALL values were roughly 200–900, while AML values were roughly 1,600–4,000. A cutoff of 1,500 classified 37 of 38 development samples correctly (97.37%).

CHRNA7 encodes the alpha-7 nicotinic acetylcholine receptor subunit and is located at chromosome 15q13.3 [14]. In this dataset, X70297 was the highest-weighted probe in both linear models. Mean expression was approximately 721 for AML and 220 for ALL. A cutoff of 340 classified 36 of 38

development samples correctly (94.74%). Previous gene-selection analyses of the Golub data have also reported X70297 among informative probes [10].

Table 6: Exploratory single-probe thresholds on the 38-sample development set.

Probe	Annotation	ALL mean	AML mean	Rule for AML	Development accuracy
X95735	<i>ZYX</i> (zyxin)	350	3,024	Expression > 1,500	37/38 (97.37%)
X70297	<i>CHRNA7</i>	220	721	Expression > 340	36/38 (94.74%)

The threshold findings should be interpreted as hypothesis-generating. First, the cutoff was optimized on the same development data used to report accuracy. Second, microarray probes can be affected by normalization, batch effects, cross-hybridization, and platform-specific scaling. Third, expression differences may reflect cell lineage or disease state without implying that a gene causes leukemia. The appropriate next step is to apply the fixed thresholds to an independent dataset and examine whether the separation remains stable.

5 Discussion

5.1 Principal findings

This comparative analysis produced three main findings. First, gene-expression profiles strongly separated AML from ALL in the Golub benchmark. Two models correctly classified more than 90% of the independent samples. Second, identical overall accuracy did not imply identical clinical behavior. The decision tree prioritized AML sensitivity, while the SVM made more conservative AML predictions with no false positives. Third, logistic regression and the SVM produced substantially overlapping feature rankings, with X70297 and X95735 emerging as repeated candidates.

The decision tree was the strongest model when missing an AML case was treated as the most

serious error. It detected 13 of 14 AML samples, compared with 11 for the SVM and 8 for logistic regression. The SVM may nevertheless be preferred in a setting where false-positive AML predictions carry a high cost, because every test sample predicted as AML by the SVM was actually AML. In practice, model selection should therefore be driven by the relative consequences of different errors rather than by accuracy alone.

The agreement between linear-model rankings is encouraging. Sixteen shared probes indicate that the results are not entirely dependent on one algorithm. The presence of established myeloid-associated probes such as *CD33* and *MPO* also supports biological plausibility. Zyxin is especially interesting because earlier analyses of the same benchmark identified it as a strong discriminator [8, 9]. This replication across methods is more persuasive than a feature appearing in only one fitted model.

5.2 Scientific and clinical interpretation

The analysis demonstrates a computational principle: cancer subtypes that can appear similar at a broad morphological level may show distinct multigene expression patterns. A classifier can combine thousands of individually noisy measurements into a useful decision boundary. This was the central contribution of the original Golub study and remains an important idea in modern molecular oncology [4].

However, high predictive weight is not equivalent to biological causation or clinical utility. A probe may be useful because it marks a cell lineage, correlates with another causal process, or captures a technical feature of the dataset. Before a gene can be proposed as a biomarker, its measurement must be replicated across cohorts and platforms, its incremental value beyond standard tests must be demonstrated, and its clinical operating characteristics must be evaluated prospectively. The current results are best described as computational evidence that these probes carry discriminative information in this benchmark dataset.

5.3 Limitations

Several limitations restrict the strength of the conclusions.

- **Small sample size.** Only 38 samples were available for model development and 34 for testing. A difference of one or two patients changes the reported percentages substantially, as reflected in the wide confidence intervals.
- **Extreme dimensionality.** With 7,129 probes and 38 development samples, many alternative models can fit the data. Regularization and independent testing reduce this risk but do not eliminate it.
- **Potential feature-selection bias.** Gene ranking and cutoff optimization can be biased when the same observations are reused for selection and evaluation. This issue is well documented in microarray studies [11].
- **Single historical platform.** The data were generated using a microarray platform from the late 1990s. Modern RNA sequencing, updated annotations, and improved laboratory protocols may produce different feature rankings.
- **Limited biological heterogeneity.** Contemporary classifications recognize multiple molecular subtypes within both AML and ALL. A binary benchmark cannot represent the full diversity encountered in clinical practice.
- **Incomplete reproducibility details.** The source draft did not record the exact software versions, regularization strengths, tree-pruning settings, or random-state choices. These details should be reported in a future reproducible analysis.
- **No calibration analysis.** Accuracy and class metrics were evaluated, but the reliability of predicted probabilities was not assessed. Calibration would be necessary before probabilistic outputs could support risk communication.

- **No external validation of single-gene thresholds.** The zyxin and *CHRNA7* cutoffs were optimized on the development cohort and should not be treated as validated diagnostic rules.

5.4 Future research

A stronger follow-up study should use nested cross-validation within the development cohort for model tuning and feature selection, followed by one final evaluation on a completely untouched external cohort. Sparse logistic regression or elastic net could identify smaller, more stable signatures while accounting for groups of correlated genes. Bootstrap resampling could quantify how often each probe appears among the top features, converting a single ranking into a stability estimate.

Additional analyses should include receiver-operating-characteristic and precision-recall curves, probability calibration, sensitivity analyses across preprocessing choices, and pathway-enrichment analysis of recurring genes. Modern RNA-seq cohorts should be used to determine whether zyxin, *CHRNA7*, *CD33*, *MPO*, and other consensus features remain informative across technologies and patient populations. Finally, any proposed biomarker panel should be evaluated alongside established clinical variables to test whether it adds information beyond morphology, immunophenotyping, cytogenetics, and current molecular diagnostics.

6 Conclusion

This project compared a decision tree, logistic regression, and a linear SVM for classifying AML and ALL from gene-expression profiles. The decision tree and SVM achieved the highest test accuracy at 91.18%, while the decision tree identified the largest proportion of AML cases. Logistic regression and the SVM shared 16 of their top 20 probes, with X70297 (*CHRNA7*) ranked first in both and X95735 (*ZYX*) appearing in both lists. Exploratory thresholds based on these two probes produced high development-set accuracy, but those cutoffs require independent validation.

The broader lesson is that a useful machine-learning study must go beyond a single performance number. Confusion matrices reveal clinically meaningful error trade-offs, cross-model agreement helps evaluate feature robustness, and careful separation of training, selection, and testing is essential in high-dimensional biomedical data. The Golub dataset remains an effective demonstration of how gene-expression patterns can support cancer classification, while also illustrating the statistical caution required before computational findings can be translated into clinical practice.

References

- [1] National Cancer Institute. *Leukemia—Patient Version*. <https://www.cancer.gov/types/leukemia>
- [2] National Cancer Institute. *Acute Myeloid Leukemia Treatment (PDQ), Patient Version*. <https://www.cancer.gov/types/leukemia/patient/adult-aml-treatment-pdq>
- [3] National Cancer Institute. *Acute Lymphoblastic Leukemia Treatment (PDQ), Patient Version*. <https://www.cancer.gov/types/leukemia/patient/adult-all-treatment-pdq>
- [4] Golub, T. R., Slonim, D. K., Tamayo, P., et al. (1999). Molecular classification of cancer: Class discovery and class prediction by gene expression monitoring. *Science*, 286(5439), 531–537. <https://doi.org/10.1126/science.286.5439.531>
- [5] Dudoit, S., Fridlyand, J., and Speed, T. P. (2002). Comparison of discrimination methods for the classification of tumors using gene expression data. *Journal of the American Statistical Association*, 97(457), 77–87. <https://doi.org/10.1198/016214502753479248>
- [6] Cortes, C., and Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20, 273–297. <https://doi.org/10.1007/BF00994018>
- [7] Furey, T. S., Cristianini, N., Duffy, N., Bednarski, D. W., Schummer, M., and Haussler, D. (2000). Support vector machine classification and validation of cancer tissue samples using microarray expression data. *Bioinformatics*, 16(10), 906–914. <https://doi.org/10.1093/bioinformatics/16.10.906>
- [8] Li, W., and Yang, Y. (2002). How many genes are needed for a discriminant microarray data analysis? In S. M. Lin and K. F. Johnson (Eds.), *Methods of Microarray Data Analysis* (pp. 137–150). Kluwer Academic. https://doi.org/10.1007/978-1-4615-0873-1_11
- [9] Tang, W., Cao, H., Duan, J., and Wang, Y.-P. (2011). A compressed sensing based approach for subtyping of leukemia from gene expression data. *Journal of Bioinformatics and Computational Biology*, 9(5), 631–645. <https://doi.org/10.1142/S0219720011005689>
- [10] Mahata, P., and Mahata, K. (2007). Selecting differentially expressed genes using minimum probability of classification error. *Journal of Biomedical Informatics*, 40(6), 775–786. PMID: 17950675.

- [11] Ambroise, C., and McLachlan, G. J. (2002). Selection bias in gene extraction on the basis of microarray gene-expression data. *Proceedings of the National Academy of Sciences*, 99(10), 6562–6566. <https://doi.org/10.1073/pnas.102102699>
- [12] Hirata, H., Tatsumi, H., and Sokabe, M. (2008). Mechanical forces facilitate actin polymerization at focal adhesions in a zyxin-dependent manner. *Journal of Cell Science*, 121(17), 2795–2804. <https://doi.org/10.1242/jcs.030320>
- [13] National Center for Biotechnology Information. *ZYX zyxin [Homo sapiens]*, Gene ID 7791. <https://www.ncbi.nlm.nih.gov/gene/7791>
- [14] National Center for Biotechnology Information. *CHRNA7 cholinergic receptor nicotinic alpha 7 subunit [Homo sapiens]*, Gene ID 1139. <https://www.ncbi.nlm.nih.gov/gene/1139>
- [15] National Center for Biotechnology Information. *CD33 molecule [Homo sapiens]*, Gene ID 945. <https://www.ncbi.nlm.nih.gov/gene/945>
- [16] National Center for Biotechnology Information. *MPO myeloperoxidase [Homo sapiens]*, Gene ID 4353. <https://www.ncbi.nlm.nih.gov/gene/4353>